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Operating Systems

Provided by Sarim

Q1) Exit system call

Answer:

The exit system call is used to terminate a process in an operating system.

Q2) Wait system call

Answer:

The wait system call is used by a parent process to wait for the termination of its child processes.

Q3) Why mknod() call fail give two reason.

Answer:

The mknod() call may fail due to insufficient permissions or if the specified file already exists.

Q4) CPU schedulers

Long term

Short term

Middle term

Answer:

- Long-term CPU schedulers select processes from the job queue and decide which ones to load into memory.
- Short-term CPU schedulers select processes from the ready queue and allocate the CPU to them.

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- Medium-term CPU schedulers remove processes from memory to alleviate congestion and improve system performance.

Q5) ps command and working of

Ps-e

Ps-u

Ps-l

Answer:

The "ps" command in Linux/Unix is used to display information about processes.

- "ps -e" displays information about all running processes.
- "ps -u" displays information about processes owned by a specific user.
- "ps -l" displays detailed information about processes in a long format.

Q6) Process in Unix/ Linux

Answer:

Processes in Unix/Linux are instances of running programs, each with its own process ID (PID), memory space, and state.

Q7) Input command in Linux/ Unix

Answer:

Input commands in Linux/Unix are entered via the command line interface, allowing users to interact with the operating system and execute various tasks.

Q8) one-to-many mapping

Answer:

One-to-many mapping refers to a relationship where one entity can be associated with multiple other entities.

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Q9) File descriptors for file created with reason

Answer:

File descriptors for files are created to provide a means for processes to access and manipulate files.

Q10) CPU scheduling long short and medium term

Answer:

CPU scheduling involves long-term scheduling (selecting processes from the job queue), short-term scheduling (allocating CPU to ready processes), and medium-term scheduling (removing processes from memory).

Q11) one to one disadvantage

Answer:

The disadvantage of one-to-one mapping in operating systems is that it requires a large amount of memory to maintain a separate kernel thread for each user thread.

Q12) Fork parent or child me kon sy process hoty 5 define

Answer:

In the fork system call in Unix/Linux, both the parent and child processes execute concurrently. The parent process gets the process ID of the child, while the child process gets a process ID of 0.

Q13) Differentiate between absolute and relative path?

Answer:

- An absolute path specifies the exact location of a file or directory from the root directory.
- A relative path specifies the location of a file or directory relative to the current working directory.

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Q14) write the code for input redirection. command < input-file or command 0< input-file

Answer:

Input redirection in Linux/Unix is achieved using the "<" symbol followed by the input file name.

Q15) Advantages of cooperating process.

Answer:

The advantages of cooperating processes include resource sharing, improved system responsiveness, and enhanced modularity and code reuse.

Q16) Scenario were given. commands were to write A person wants to foreground the process (number 1234), job ID, He wants to background the process

Answer:

- To foreground a process with job ID 4: "fg [4]"
- To background a process: "bg []"
- To show the status of suspended processes: "jobs"

Q17) Write three functions of operating system

Answer:

Three functions of an operating system are process management, memory management, and file system management.

Q18) Preemptive or non-preemptive definition with one example

Answer:

Preemptive scheduling allows higher priority tasks to interrupt lower priority tasks, while non-preemptive scheduling completes the current task before switching to another.

Example:

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Preemptive scheduling is used in modern operating systems like Windows and Linux, where the CPU can be interrupted by higher priority tasks.

Non-preemptive scheduling is used in some real-time systems where tasks have fixed priorities.

Q19) Write 5 Unix/Linux commonly used IPC tools

Answer:

Five commonly used IPC (Interprocess Communication) tools in Unix/Linux are

I. Pipes
II. Message queues
III. Semaphores
IV. Shared memory
V. Sockets

Q20) Write 2 limitations of named and unnamed pipe

Answer:

Two limitations of named and unnamed pipes are

1. Named pipes have a limited capacity, which can lead to blocking if the buffer is full.
2. Unnamed pipes can only be used for communication between related processes, such as parent-child processes.

Q21) Write down the three main failure of folk() system call?

Answer:

Three main failures of the fork() system call are:

- Resource duplication issues.
- Failure to create a new process due to insufficient resources.
- Incorrect execution due to errors in the fork() implementation.

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Q22) The ps is used to display status of processes. The -e, -u, -l is also used with ps command. Write the use of the following command.

a). ps -e

b). ps -u

c). ps -l

Answer:

a) "ps -e" displays information about all running processes.

b) "ps -u" displays information about processes owned by a specific user.

c) "ps -l" displays detailed information about processes in a long format.

Q23) The two process is Communicating by pipe or named pipe(FIFO). Write the process of communication through pipe. Also write syntax.

Answer:

Communication through a pipe involves one process writing data to the pipe and another process reading from it.

The syntax for creating a pipe in Unix/Linux is:

int pipe(int pipefd[2]);

Q24) Write names of five tools used for IPC in Unix operating environment.

Answer:

Five tools used for IPC in Unix operating environment are:

1. Pipes
2. Message queues
3. Semaphores
4. Shared memory
5. Sockets

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Q25) X and Y want to communicate to each other how many way they can communicate

Answer:

X and Y can communicate with each other using several methods such as pipes, message queues, sockets, shared memory, and signals.

Q26) root directory ki 3 characteristics btani thi

Answer:

1. It is the top-most directory in a file system hierarchy.
2. It serves as the parent directory for all other directories and files.
3. It has a special symbol ("/" in Unix/Linux) to represent it.

Q27) unix/ linux ki koi situation thi wha btana kn sa ipc apply ho ga

Answer:

In Unix/Linux, if multiple processes need to communicate with each other to share data or synchronize their activities, the situation may call for the use of Inter-Process Communication (IPC) mechanisms such as pipes, message queues, or shared memory.

Q28) Mutual threading

Answer:

Mutual threading refers to a situation where multiple threads share access to a resource, but each thread must wait for other threads to release the resource before it can access it.

Q29) Name three directory used in Linux operating system?

Answer:

Three directories used in Linux operating system

- /bin
- /etc
- /home

Q30) Round robin scheduling

Answer:

Round Robin scheduling is a CPU scheduling algorithm that allocates CPU time to processes in a circular manner, where each process gets a small unit of CPU time (a time slice) and then moves to the back of the ready queue.

Q31) Difference b/t single thread and multi thread

Answer:

Difference between single-threaded and multi-threaded processes:

- Single-threaded processes have only one execution path (thread) of execution, while multi-threaded processes have multiple threads of execution.

Q32) IPC ky process names

Answer:

Inter-Process Communication (IPC) mechanisms in Unix/Linux

1. Pipes
2. Message queues
3. Shared memory
4. Semaphores
5. Signals

Q33) Subject turnaround time define

Answer:

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Turnaround time in the context of operating systems refers to the total time taken for a process to execute, including both waiting time and execution time.

Q34) When CPU is in waiting state it still work hum is state ko Kya khty definition btani thi

Answer:

When the CPU is in the waiting state, it is still working but is waiting for an I/O operation to complete.

This state is known as the I/O waiting state, where the CPU is idle but other operations, such as data transfer, are being performed.

Q35) Name five tools IPC for Unix and Linux operating system

Answer:

Five tools for Interprocess Communication (IPC) in Unix and Linux are: Pipes, Message Queues, Semaphores, Shared Memory, and Sockets.

Q36) multiple system shared kerta ha resources give 3 security issue

Answer:

Three security issues with multiple systems sharing resources are

- Unauthorized access to shared resources
- Data integrity concerns
- Resource contention leading to performance degradation or denial of service

Q37) Two limitations of using of unnamed pipes or simple pipes in Unix/Linux proceed communication

Answer:

Two limitations of using unnamed pipes in Unix/Linux for interprocess communication is

1. Limited to communication between related processes (e.g., parent-child)

2. Limited capacity can lead to blocking if the buffer is full

Q38) System call can be categorized into the following groups

Answer:

System calls can be categorized into several groups such as process control, file management, device management, information maintenance, communication, and protection.

Q39) Critical solution problem and its approaches are necessary?

Answer:

Critical section problem approaches include the use of semaphores, locks, and monitors to synchronize access to shared resources among concurrent processes or threads.

Q40) Kernel bottleneck

Answer:

Kernel bottleneck occurs when the kernel becomes a performance bottleneck due to excessive resource consumption or inefficient resource management, leading to degraded system performance.

Q41) Linux commands Fg, kill, PS, bg, cp

Answer:

Linux commands include

1. "fg" for bringing a background process to the foreground
2. "kill" for terminating processes
3. "ps" for displaying information about running processes
4. "bg" for running processes in the background
5. "cp" for copying files.

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Q42) Kernel level thread

Answer:

Kernel-level threads are managed by the operating system kernel, allowing multiple threads to execute concurrently within the same process context.

Q43) Turn time or responsive time differences

Answer:

Turnaround time refers to the total time taken by a process to complete execution, while response time refers to the time taken for a system to respond to a user request.

The difference lies in the inclusion of waiting time in turnaround time.

Q44) Semaphore 2 atomic k name with code

Answer:

Two atomic semaphore functions are wait() and signal(), typically implemented as atomic operations to avoid race conditions in concurrent programming.

Q45) Short term scheduler

Answer:

Short-term scheduler, also known as CPU scheduler, selects which process from the ready queue will be executed next by the CPU.

Q46) Preemptive and non-preemptive SJF

Answer:

Preemptive Shortest Job First (SJF) scheduling involves preempting the currently running process if a shorter job arrives with higher priority.

Non-preemptive SJF scheduling completes the current job before switching to another.

Q47) Hardware solution of critical section me two variables code

Answer:

Hardware solution of the critical section problem typically involves using hardware instructions like TestAndSet or CompareAndSwap to ensure mutual exclusion among concurrent processes accessing shared resources.

Q48) process me Jo stages hoti hain 5

Answer:

The five stages of a process are

New, Ready, Running, Waiting, and Terminated

Q49) BSD socket in tools

Answer:

BSD sockets are a set of APIs used for network communication in Unix-like operating systems such as Linux and BSD.

They provide a standard interface for networking operations and are commonly used for TCP/IP communication.

Q50) Errors batane thy interrupts, trap and signal.

Answer:

Interrupts occur in response to hardware signals, traps are synchronous events caused by software instructions, and signals are asynchronous events sent by processes or the kernel to other processes.

Q51) System call functions

Answer:

System call functions include fork(), exec(), open(), read(), write(), and close().

Q52) Sort the scheduling arranges the high to low.

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Answer:

Sorting scheduling arrangements from high to low priority

Real-time scheduling, Short-term scheduling, Medium-term scheduling, Long-term scheduling.

Q53) List five tool use in IPC Unix operating system environment.

Answer:

Five tools used in Interprocess Communication (IPC) in Unix operating system environment are: Pipes, Message Queues, Semaphores, Shared Memory, and Sockets.

Q54) List many operations are use in atomic operation samephore wrote name and code.

Answer:

Operations used in atomic operations with semaphores: wait() and signal().

Example code:

```
// Semaphore wait operation

wait(S);

// Semaphore signal operation

signal(S);
```

Q55) Why fork system call fails state three reasons.

Answer:

Fork system call may fail due to insufficient memory resources, reaching the maximum number of processes allowed, or if the process lacks permission to fork a new process.

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